

SaResQ — Presentation Draft

Slide-by-slide script: what to put on each slide, and what to say

FOR	DATE	SLIDES	RUNTIME
Smart India Hackathon	11 September 2026	12 core + 3 reserve	~8 minutes

How to use this. Each block below is one slide. **ON SLIDE** is the text that goes on screen — keep it short, it is not a document. **SAY** is your spoken line. **VISUAL** is what to build. Every number here is measured and traceable to the technical report.

The narrative arc: drone search is already possible — but a human has to watch the video. That human is the bottleneck. We put the watching onboard. Doing that on a ₹4,000 computer is the hard engineering, and we have the measurements to prove we did it.

Act I — The problem

01 Title10 SECONDS

ON SLIDE

SaResQ — onboard AI that finds survivors, so nobody has to watch the screen

Thermal + visual search payload for disaster response · Team name · SIH 2026

VISUAL

One photograph of the actual printed payload with the Pi fitted. A real object beats a render — it proves the thing exists.

02 The windowSTAKES

ON SLIDE

After a building collapse, survival falls sharply with every hour trapped.

Search teams are slow, scarce, and at risk. Drones can cover ground fast.

Say: "Everyone in this room already knows drones can search a collapse site. That problem is solved. The problem nobody has solved is what happens to the video."

VISUAL

One image of a collapse site. No bullet points. Let the photograph carry it.

Insert a sourced survival-rate statistic here — use one you can cite, since a judge may ask. Don't invent a number.

file:///Users/afzalamanullah/saresq/docs/deck_draft.html

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03 The real bottleneck

THE INSIGHT — MOST IMPORTANT SLIDE

ON SLIDE

The drone isn't the bottleneck. The person watching the feed is.

- Waypoint flight has been solved for a decade
- A survivor is a **2.6-pixel** warm smudge on a low-resolution thermal feed at 4 frames a second
- Human visual search degrades sharply with time on task
- An operator cannot rewind, cannot combine three weak sightings, cannot attach a GPS fix to a hunch

Say: "We are not automating the flying. We are automating the *looking* — because that is the part that fails."

VISUAL

An actual 32×24 thermal frame with a person in it, blown up. Let the audience try to spot them. Then reveal the box. This lands hard — it makes the difficulty physical rather than asserted.

Act II — The solution

04 What we built

THE REVEAL

ON SLIDE

A 340-gram payload that watches for you — onboard, in real time.

₹14k

total bill of materials

340 g

fits a 450 mm quadcopter

0 wires

to the flight controller

Thermal camera · Pi Camera · Raspberry Pi 4 · IMU · GPS · own battery. Bolts or straps to a drone you already own.

Say: "It carries its own power, its own GPS, its own everything. It never touches the flight stack — so it cannot crash your drone, and fitting it is a mechanical job, not an avionics project."

05 How it works

ARCHITECTURE — KEEP IT TO 40 SECONDS

ON SLIDE

Look cheaply everywhere. Look expensively almost nowhere.

STAGE	DOES	COST
1. Thermal gate	Flags warm <i>or cold</i> anomalies in a 32×24 frame. Pure statistics, no AI.	~0 ms
2. Detector	AI looks <i>only</i> at the 2–6 flagged patches	28 ms each
3. Scene classifier	Fire? Flood? Collapse? — context, once a second	33 ms
4. Fusion	Combines evidence across repeated passes into one probability + GPS fix	<0.1 ms

Say: "A full frame is 409,600 pixels. A patch is 25,600. We do about a sixteenth of the work a normal detector would — that is the entire reason this runs on a ₹4,000 computer instead of a laptop."

VISUAL

A horizontal 4-box pipeline, each box labelled with its cost. Animate left to right if the format allows.

06 Cold matters too

DEPTH — SHOWS YOU THOUGHT PAST THE OBVIOUS

ON SLIDE

We look for cold anomalies as well as warm ones.

- A body in water reads **colder** than its surroundings
- A void beneath rubble reads colder than sun-baked concrete
- In direct afternoon sun, debris can be **hotter than a human being** — a "find hot things" system fails exactly then

Say: "A naive thermal detector looks for hot spots. In Kolkata sun, the rubble is the hot spot. So our gate is two-sided — and we recommend flying at dawn, dusk or night. That's a physics constraint, not a software one."

This slide is optional if time is tight — but it is the kind of detail that separates a project that was thought about from one that was assembled.

Act III — Proof

07 Does it find people?

HEADLINE RESULT

ON SLIDE

85% on the dataset it was built for. 54% on a completely different one.

DATASET	WHAT IT IS	PERSON	AP50
HIT-UAV	Aerial thermal, overhead	0.850	
RGBTDronePerson	Night, urban, dense, oblique	0.544	
Scene classifier	AIDER — fire / flood / collapse	95.2%	

Say: "We quote two datasets on purpose. A single number always invites the question 'did it just memorise that dataset?' Two independently collected datasets answer it before you ask."

If asked "is 85% good?" — "For a 3-megabyte model on a Raspberry Pi, yes. It isn't state of the art; a model fifty times bigger would beat it. The constraint is the point."

08 Does it actually run?

THE CREDIBILITY SLIDE

ON SLIDE

Measured on the Raspberry Pi. Not estimated.

28 ms

per patch, on the Pi

177 ms

worst case, of a 250 ms budget

95 MB

RAM used, of 905 available

29% headroom at the worst case. 43 °C, no thermal throttling, running off the payload's own power bank.

Say: "Here is the number that matters. A conventional whole-frame detector would need about 320 milliseconds on this hardware — that's over the entire frame budget before it does anything else. Ours needs 177. The architecture isn't a preference, it's the only thing that fits."

VISUAL

A screenshot of the actual terminal output from the Pi benchmark. Raw evidence is more persuasive than a styled chart.

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09 How we know it's right

YOUR DIFFERENTIATOR — DO NOT CUT THIS

ON SLIDE

We measured the obvious choices — and two of them were wrong.

DECISION	THE OBVIOUS CHOICE	WHAT MEASUREMENT SHOWED
Detector head	Add a high-resolution layer for small objects	Lost in every size bin at 2× the compute
Compression	Standard 8-bit quantisation	Silently ran at 32% accuracy — no error, plausible output

Say: "The second one is the one I'd want you to remember. Our deployed model was scoring a third of what we thought. It loaded fine, it ran fast, it drew sensible-looking boxes. Nothing failed. We only found it because we built a test that compares the model we ship against the model we quote — and we'd have found out on stage instead."

This is the strongest slide in the deck. It demonstrates engineering judgement, not just a working demo — and every judge has seen projects that would have shipped that bug.

Act IV — Fit and future**10 It fits your drone**

FEASIBILITY

ON SLIDE

**Bolts to any 450 mm-class quadcopter.
No firmware change. No re-certification.**

- 4 × M3 into a standard vibration-damping plate, or 2 straps through the built-in slots
- Own battery, own GPS, own IMU — **zero electrical connection to the flight controller**
- Flat layout keeps the centre of gravity low; prints with no supports
- GPS antenna on a separate mast — a Pi radiates across the GPS band at 1575 MHz

Say: "Because it never touches the flight stack, it cannot brown it out and cannot cause a flyaway. Fitting it to a new airframe is a bolt-on, not an integration project."

VISUAL

The CAD render or a photo of the printed tray with the mounting holes called out.

11 What autonomy this adds

THE ASK BEHIND THE ASK

ON SLIDE

Today: the drone sees for itself. Next: it decides where to look.

LEVEL	MEANING	STATUS
Lo	Pilot flies, human watches the feed	Today's practice
L1	Payload detects, fuses across passes, geo-tags — unsupervised	Built & running
L2	Payload re-tasks the aircraft: orbit a hit, densify the search	One cable away

The Pi's serial port is already free for MAVLink. We have **deliberately** not connected it.

Say: "The moment the payload can command the aircraft, every false alarm becomes an unplanned manoeuvre. That step comes after field validation, not before. Right now the electrical independence is exactly what lets this fly today with no safety case."

Restraint reads as maturity. Judges probe for people who overclaim autonomy — this pre-empts it.

12 Close

LAND IT

ON SLIDE

₹14,000 of parts. 340 grams. Finds 85% of people, onboard, in real time.

Works on a drone you already own.

Say: "Every number we have shown you was measured on this hardware — including the ones that told us we were wrong. That's the project."

Reserve slides — for Q&A

R1 What we know doesn't work yet

ANSWER THIS BEFORE THEY ASK

ON SLIDE

Honest limits

- **Never flown.** Benchmarked on the board, not in the air.
- **The evidence-fusion stage is weak** (0.67 AUC) — 5 of its 18 inputs need sensors mounted and flying.
- **No daytime visible detector.** Thermal-only today; the dataset we had for visible turned out to be night-time imagery with no usable signal.
- **Search-coverage figure withheld** — the only measurement available is out-of-domain, and publishing it would overstate coverage roughly 5×.

Say: "We'd rather tell you the weak parts than have you find them."

R2 Best operating altitude

LIKELY QUESTION

ON SLIDE

20–30 m. The limit is the thermal sensor, not the AI.

ALTITUDE	THERMAL PIXELS ON A PRONE ADULT	SWATH
20 m	2.1	21 m
25 m	1.4	26 m
40 m	0.5 – contrast halves	42 m

Below one thermal pixel the person's heat averages with the cold ground around them, and apparent contrast falls as $1/\text{height}^2$. The detector barely degrades over this range — the sensor does.

R3 Why not just use a bigger model / a Jetson?

LIKELY CHALLENGE

ON SLIDE

Because the constraint *is* the contribution.

- A Jetson is 5–10× the cost and draws far more power — which is flight time
- Anyone can run a large model on a large computer; that isn't deployable at scale in a disaster response budget
- Our model is **3.2 MB** and uses **95 MB** of RAM — a state relief agency can buy fifty of these

Say: "The engineering question isn't 'can AI find people in thermal images' — it's 'can it do that on hardware a state disaster response force can actually afford to buy in quantity'."

Build notes

- **Slides 3, 8 and 9 are the ones that win it.** If you have to cut, cut 6 and 12 first.
- **Put a real artefact on screen early** — the printed payload on slide 1, a real thermal frame on slide 3, the raw Pi terminal output on slide 8.
- **Don't animate numbers.** Let them sit still long enough to be read.

- **The video** should show, in order: the payload being fitted to a drone; the thermal feed with the gate firing; the detector confirming; the map pin appearing. Even a bench demo with someone walking past the sensor carries this.
- Every figure traces to `docs/SaResQ_Model_Architecture_Report.pdf` . If a judge pushes on any number, it is defensible.